

Markscheme

Mathematics: analysis and approaches

Practice question bank

154. (a)

One root of a quadratic equation with real coefficients is $z = 3 - 2i$.

Find the sum and the product of the two roots of the equation.

the other root is the conjugate $3 + 2i$ (M1)

sum = $2\operatorname{Re}(z) = 6$; product = $|z|^2 = 9 + 4 = 13$ A1

[3 marks]